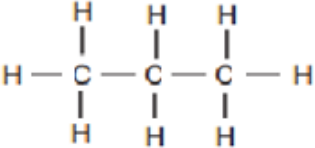
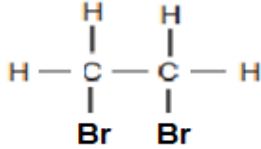
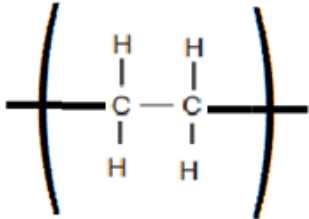
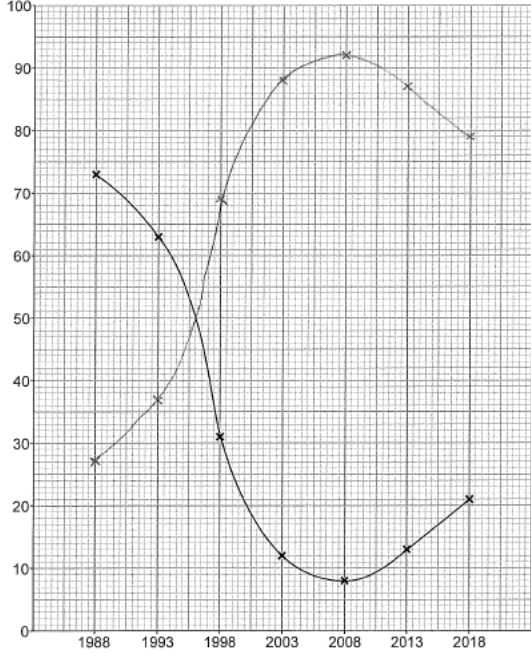


COMMON QUESTIONS

| Question |     |       |  | Marking details                                                                                                                                                                     | Marks available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7/1      | (a) | (i)   |  | <b>D and E</b> <b>both</b> needed, either order                                                                                                                                     |                 | 1   |     | 1     |       |      |
|          |     | (ii)  |  | propene                                                                                                                                                                             | 1               |     |     | 1     |       |      |
|          |     | (iii) |  |                                                                                                    | 1               |     |     | 1     |       |      |
|          | (b) | (i)   |  |  (1)<br><br> (1) |                 | 2   |     | 2     |       |      |
|          |     | (ii)  |  | decolourises / goes colourless<br><br>neutral answers – goes clear / changes colour                                                                                                 | 1               |     |     | 1     |       | 1    |

| Question |     |       |  | Marking details                                                                                                                                       | Marks available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                       | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |     | (iii) |  | (addition) polymerisation                                                                                                                             | 1               |     |     | 1     |       |      |
|          | (c) | (i)   |  | 0.15 (2)<br><br>if answer incorrect award (1) for any of following<br><br>$\frac{12}{100} \times 1.25$<br><br>$0.0125 \times 12$<br><br>$15 / 150000$ |                 | 2   |     | 2     | 2     |      |

| Question |                       |                       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |                       |                       |       |       |      |      |    |    |      |    |    |      |    |    |      |   |    |      |    |    |      |    |    |  |   |  |   |   |  |
|----------|-----------------------|-----------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------|-----------------------|-------|-------|------|------|----|----|------|----|----|------|----|----|------|---|----|------|----|----|------|----|----|--|---|--|---|---|--|
|          |                       |                       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2                   | AO3                   | Total | Maths | Prac |      |    |    |      |    |    |      |    |    |      |   |    |      |    |    |      |    |    |  |   |  |   |   |  |
|          |                       | (ii)                  |  | <div><table><caption>Data points from the graph</caption><tr><th>Year</th><th>Series 1 (Decreasing)</th><th>Series 2 (Increasing)</th></tr><tr><td>1988</td><td>73</td><td>27</td></tr><tr><td>1993</td><td>63</td><td>37</td></tr><tr><td>1998</td><td>31</td><td>69</td></tr><tr><td>2003</td><td>12</td><td>88</td></tr><tr><td>2008</td><td>8</td><td>92</td></tr><tr><td>2013</td><td>13</td><td>87</td></tr><tr><td>2018</td><td>21</td><td>79</td></tr></table></div> <p>award (2) for <b>all</b> points plotted correctly – tolerance <math>\pm 1</math> square</p> <p>award (1) for any <b>four</b> correct points</p> <p>award (1) for curved line</p> <p>do not accept - point to point line</p> | Year            | Series 1 (Decreasing) | Series 2 (Increasing) | 1988  | 73    | 27   | 1993 | 63 | 37 | 1998 | 31 | 69 | 2003 | 12 | 88 | 2008 | 8 | 92 | 2013 | 13 | 87 | 2018 | 21 | 79 |  | 3 |  | 3 | 3 |  |
| Year     | Series 1 (Decreasing) | Series 2 (Increasing) |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                       |                       |       |       |      |      |    |    |      |    |    |      |    |    |      |   |    |      |    |    |      |    |    |  |   |  |   |   |  |
| 1988     | 73                    | 27                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                       |                       |       |       |      |      |    |    |      |    |    |      |    |    |      |   |    |      |    |    |      |    |    |  |   |  |   |   |  |
| 1993     | 63                    | 37                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                       |                       |       |       |      |      |    |    |      |    |    |      |    |    |      |   |    |      |    |    |      |    |    |  |   |  |   |   |  |
| 1998     | 31                    | 69                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                       |                       |       |       |      |      |    |    |      |    |    |      |    |    |      |   |    |      |    |    |      |    |    |  |   |  |   |   |  |
| 2003     | 12                    | 88                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                       |                       |       |       |      |      |    |    |      |    |    |      |    |    |      |   |    |      |    |    |      |    |    |  |   |  |   |   |  |
| 2008     | 8                     | 92                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                       |                       |       |       |      |      |    |    |      |    |    |      |    |    |      |   |    |      |    |    |      |    |    |  |   |  |   |   |  |
| 2013     | 13                    | 87                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                       |                       |       |       |      |      |    |    |      |    |    |      |    |    |      |   |    |      |    |    |      |    |    |  |   |  |   |   |  |
| 2018     | 21                    | 79                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                       |                       |       |       |      |      |    |    |      |    |    |      |    |    |      |   |    |      |    |    |      |    |    |  |   |  |   |   |  |

| Question |  |       |    | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks available |     |     |       |       |      |
|----------|--|-------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |       |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  | (iii) | I  | 1996<br><br>accept 1995 / 1997<br>accept range 1995-1996 / 1996-1997                                                                                                                                                                                                                                                                                                                                                                                                                             |                 | 1   |     | 1     |       | 1    |
|          |  |       | II | 2 : 8    1 : 4    4 : 1    8 : 2    20 : 80                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |     | 1   | 1     | 1     |      |
|          |  | (iv)  |    | award (1) for any of following actions<br><br>people <b>use</b> bag for life / alternative bags<br>people <b>reuse</b> bags<br>supermarkets <b>charge</b> for bags / <b>stop giving</b> free plastic bags<br>government <b>introduced</b> a charge<br>people have <b>become more aware</b> of issues to do with plastic waste<br>people <b>want to reduce</b> amount of plastic going to landfill / incineration<br>people <b>want to reduce</b> the amount of plastic pollution / microplastics |                 |     | 1   | 1     |       |      |
|          |  |       |    | Question 7/1 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4               | 9   | 2   | 15    | 6     | 2    |